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WASTE COMPOSTER AND WEIGHING METHOD THEREOF

Abstract

A waste composter includes a barrel body assembly, a driving assembly, a base, and a weighing assembly. The barrel body assembly includes a barrel body and a stirring member disposed in the barrel body. The driving assembly is connected to the barrel body and the stirring member. The base includes a bottom plate and a plurality of supporting bodies which are disposed around a side of the bottom plate to form an accommodation space. The driving assembly is accommodated in the accommodation space, and the barrel body is disposed above the plurality of supporting bodies. The weighing assembly is disposed between the barrel body and the plurality of supporting bodies and is configured to weigh a total mass of the barrel body assembly, the driving assembly, and the waste in the barrel body. A weighing method of the waste composter is also disclosed.

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Background/Summary

FIELD

[0001] The subject matter herein generally relates to the field of waste disposal, in particular to a waste composter and a weighing method thereof.

BACKGROUND

[0002] Food waste composters are devices widely used in homes and business locations and are used to reduce a volume of food waste and convert the food waste through physical or biological methods (such as fermentation) for more environmentally friendly disposal. The design and operation of waste composters are critical to food waste management, especially in urban environments. An effective treatment of food waste can significantly reduce the total volume of the waste, reduce reliance on landfills, and potentially generate usable by-products such as compost.

[0003] However, the food waste composters currently on the market are not equipped with weighing devices, which causes several problems. Firstly, the lack of weighing devices means that users cannot know exactly how much waste has been disposed of, which is an obvious shortcoming for business locations that need to accurately calculate the cost or efficiency of waste disposal. Secondly, the lack of weighing function also affects the efficiency of biological treatment methods such as fermentation, because these methods usually require a certain amount of waste to achieve optimal treatment results. Finally, the lack of the weighing devices also prevents the user from being alerted in time when the device is full load or overloaded, which may affect the service life of the device or even cause damage to the device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is an overall schematic diagram of a waste composter provided by an embodiment of the present disclosure.

[0005] FIG. 2 is an exploded view of a barrel body assembly, driving assembly, base, and weighing assembly of the waste composter of FIG. 1.

[0006] FIG. 3 is a schematic diagram, from another angle of view, of the barrel body assembly, driving assembly, base, and weighing assembly of FIG. 2.

[0007] FIG. 4 is a schematic diagram of a cross section of the waste composter along a line IV-IV of FIG. 1.

[0008] FIG. 5 is a schematic diagram of a stirring member and a second connecting member of the waste composter of FIG. 1.

[0009] FIG. 6 is a logic control diagram of the waste composter of FIG. 1.

[0010] FIG. 7 is a schematic diagram of a position state B, a second position state C, and a third position state D provided by an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0011] The technical solutions of the present disclosure will be described in detail below with reference to the drawings and embodiments of the present disclosure. Apparently, the described exemplary embodiments are part of embodiments of the present disclosure, not all of the embodiments.

[0012] It should be noted that when an element is considered to be “disposed” on another element, the element may be directly disposed on the other element, or there may be an intermediate element.

[0013] Referring to FIG. 1 and FIG. 2, a waste composter **100** of an embodiment of the present disclosure is illustrated. The waste composter **100** includes a barrel body assembly **10**, a driving

assembly **20**, a base **30**, and a weighing assembly **40**. The barrel body assembly **10**, the driving assembly **20**, and the weighing assembly **40** are all disposed on the base **30**. The driving assembly **20** is connected to an exterior of the barrel body assembly **10**. The weighing assembly **40** is disposed between the barrel body assembly **10** and the base **30**.

[0014] Referring to FIG. 2, the barrel body assembly **10** includes a barrel body **11** and a stirring member **12**. The stirring member **12** is rotatably installed at a bottom of the barrel body **11**. The barrel body **11** is used to receive waste, such as kitchen waste. The driving assembly **20** is drivingly connected to the stirring member **12** to drive the stirring member **12** to rotate, thereby stirring the waste contained in the barrel body **11**.

[0015] The base **30** includes a bottom plate **31** and a plurality of supporting bodies **32**. The plurality of supporting bodies **32** are arranged around one side of the bottom plate **31** to form a receiving space **33**. The driving assembly **20** is located in the receiving space **33**, and the barrel body **11** is disposed above the plurality of supporting bodies **32** in a vertical direction A.

[0016] The weighing assembly **40** is disposed between each supporting body **32** and the barrel body **11**. A total mass of the barrel body assembly **10** and the driving assembly **20** is defined as a first mass $W1$, and a total mass of the barrel body assembly **10**, the driving assembly **20** and the waste contained in the barrel body **11** is defined as a second mass $W2$. The weighing assembly **40** is used to collect the first mass $W1$ and the second mass $W2$, and the weighing assembly **40** can obtain a mass $W3$ of the waste based on the difference between these two masses, that is, $W3 = W2 - W1$. A calculation process is completed through a second electronic control assembly **62** (shown in FIG. 4 and FIG. 6).

[0017] When in use, a user can adaptively adjust a stirring speed of the stirring member **12** according to the mass of the waste, thereby helping to improve processing efficiency, reducing processing time and cost, and ensuring that the waste is fully and effectively processed. Secondly, the user can better control the mass of added microorganisms based on the mass of the waste to ensure optimal conditions during a biological treatment process, thus helping to improve the efficiency of biological treatments such as fermentation and produce more useful by-products such as compost. All in all, arrangement of the waste composter **100** having a weighing function allows the user to easily understand operation status of the waste composter **100**, which helps to improve the user experience and allows the user to manage and operate the waste composter more conveniently.

[0018] In the waste composter **100** provided by the present disclosure, the weighing assembly **40** is disposed between the barrel body assembly **10** and the supporting bodies **32** and is connected to the barrel body assembly **10** and the driving assembly **20**, so that the weighing assembly **40** can measure the total mass of the barrel body assembly **10** and the driving assembly **20**, as well as the total mass of the barrel body assembly **10**, the driving assembly **20**, and the waste accommodated in the barrel body assembly **10**. In this way, the net weight of the waste can be obtained by weighing twice, the measurement process is more accurate, and the waste composter **100** has a simple structure.

[0019] Referring to FIG. 2 and FIG. 3, in the embodiment, the barrel body assembly **10** further includes a substrate **13** and a plurality of latch portions **14**. The barrel body **11** is substantially cylindric and includes a bottom plate **111** and a side plate **112**. The side plate **112** is connected to the bottom plate **111** and disposed around a periphery of the bottom plate **111** to form an accommodation space **113**. The accommodation space **113** is used to accommodate the waste. The bottom plate **111** extends away from the side plate **112** to form the plurality of latch portions **14**. The substrate **13** is provided with a plurality of clamping grooves **141** penetrating the substrate **13** and corresponding to the plurality of latch portions **14** respectively, and each latch portion **14** is inserted into a corresponding clamping groove **141**. In this way, the substrate **13** and the barrel body **11** are connected. The weighing assembly **40** is disposed between the substrate **13** and the plurality of supporting bodies **32**. In other embodiments of the present disclosure, the clamping

grooves **141** are defined on the bottom plate **111**, and the latching portions **14** are defined on the substrate **13**.

[0020] In the embodiment, the substrate **13** is substantially square and includes a main body portion **131** and a plurality of extension portions **132**. The plurality of extension portions **132** protrude from an edge of the main body portion **131**. Each supporting body **32** is provided corresponding to one extension portion **132**. The bottom plate **111** of the barrel body **11** is disposed corresponding to the main body portion **131**. The clamping groove **141** is disposed on a connection between the main body portion **131** and the extension portion **132**. The weighing assembly **40** is disposed between the extension portions **132** and the supporting bodies **32**. The weighing assembly **40** is used to measure a total mass $W4$ of the substrate **13**, the latch portion **14**, the barrel body **11**, and the driving assembly **20**, as well as a total mass $W5$ of the substrate **13**, the latch portion **14**, the barrel body **11**, the driving assembly **20**, and the waste in the barrel body **11**. The weighing assembly **40** can obtain the mass $W3$ of the waste based on the difference between these two masses, that is, $W3=W5-W4$.

[0021] In this embodiment, the base **30** further includes a plurality of directional columns **34**, and the plurality of directional columns **34** are disposed on a side of the supporting bodies **32** facing the extension portions **132**. The plurality of directional columns **34** and the weighing assembly **40** are arranged in a staggered manner. The plurality of directional columns **34** extend toward the barrel body **11** in the vertical direction A. The extension portions **132** defines a plurality of directional holes **35** corresponding to the directional columns **34**, respectively. Each directional column **34** can movably pass through a corresponding directional hole **35**. When in use, the directional columns **34** and the directional holes **35** are used to limit the slight movement of the barrel body **11** in the vertical direction A and to avoid movement in other directions, which is beneficial to the accuracy of weighing of the weighing assembly **40**.

[0022] In the embodiment, the number of the directional columns **34** is four, and the number of the directional holes **35** is four. The number of the supporting bodies **32** is four, and the number of the extending portions **132** is four. The four supporting bodies **32** are symmetrically distributed, and the four directional pillars **34** are also symmetrically distributed. In other embodiments of the present disclosure, the number of the directional columns **34**, of the directional holes **35**, of the supporting bodies **32**, and of the extension portions **132** may be two, three, five, etc.

[0023] Referring to FIG. 2, FIG. 4, and FIG. 5, in the embodiment, the stirring member **12** includes a plurality of stirring paddles **121**, a propeller hub **122**, and a first sleeve connection member **123**. One end of each of the plurality of stirring paddles **121** is connected to a periphery of the propeller hub **122**. The first sleeve connection member **123** is disposed on one side of the propeller hub **122**. The driving assembly **20** includes a motor **21**, a speed reducer **22**, and a second sleeve connection member **23**. The motor **21** is connected to the speed reducer **22**, and the speed reducer **22** is connected to the second sleeve connection member **23**. The first sleeve connection member **123** is sleeved on the second sleeve connection member **23**. In this way, the stirring member **12** is drivingly connected to the driving assembly **20**. The speed reducer **22** is used to reduce the output speed of the motor **21** and to increase the output torque, so that the plurality of stirring paddles **121** can smoothly stir the waste. The speed reducer **22** is connected to the main body portion **131** through a fixing member (e.g., a screw), and the motor **21** is connected to the main body portion **131** through another fixing member (e.g., a screw). In this way, the driving assembly **20** and the barrel body assembly **10** are connected as a whole.

[0024] Referring to FIG. 5, in the embodiment, the first sleeve connection member **123** is substantially bowl-shaped, and the first sleeve connection member **123** includes a sleeve portion **123a** and a plurality of protrusions **123b**. The plurality of protrusions **123b** extend in the vertical direction A and are disposed in the sleeve portion **123a** in a spaced manner. The second sleeve member **23** includes a core shaft **231** and a plurality of grooves **232** defined on an outer surface of the core shaft **231** and spaced from each other. Each of the plurality of protrusions **123b** can insert

into one of the plurality of grooves **232**. In this way, the first sleeve member **123** is sleeved on the second sleeve member **23**.

[0025] Referring to FIG. **4** and FIG. **6**, in the embodiment, the weighing assembly **40** includes a first electronic control assembly **41**, a plurality of pressure sensors **43**, and a second electronic control assembly **62**. The weighing assembly **40** is configured to be communicatively connected to a mobile terminal **42**, and the mobile terminal **42** is configured to received information transmitted by the weighing assembly **40**. The second electronic control assembly **62** is electrically connected to the first electronic control assembly **41**, and the plurality of pressure sensors **43** are disposed between the plurality of extension portions **132** and the plurality of supporting bodies **32** respectively. The first electronic control assembly **41** is electrically connected to the plurality of pressure sensors **43** to supply power to the plurality of pressure sensors **43**. The plurality of pressure sensors **43** are electrically connected to the second electronic control assembly **62**, and the second electronic control assembly **62** is configured to communicatively connected to the mobile terminal **42**. The second electronic control assembly **62** is used to obtain the first mass W1 and the second mass W2 based on detection data of the plurality of the pressure sensors **43**, and to calculate the mass W3 based on the first mass W1 and the second mass W2. The mobile terminal **42** is used to display the mass W3 of the waste. In other embodiments of the present disclosure, the weighing assembly **40** further includes a wireless transmission module, which is used to transmit weight information to be displayed by the mobile terminal **42** to the mobile terminal **42**, such as a smartphone, etc. The number of the plurality of pressure sensors **43** is four, and each pressure sensor **43** is disposed between a corresponding supporting body **32** and a corresponding extension portion **132**. The second electronic control assembly **62** is electrically connected to the four pressure sensors **43** and calculates the first mass W1 or the second mass W2 by summing mass measured by four pressure sensors **43**. The second electronic control assembly **62** is one of a central processing unit, a digital signal processor, and an application-specific integrated circuit. The pressure sensors **43** are one of strain gauge pressure sensors, piezoresistive pressure sensors, and capacitive pressure sensors. It can be understood that in other embodiments of the present disclosure, the first mass W1 is stored in the second electronic control assembly **62**.

[0026] In the embodiment, the first electronic control assembly **41** is further electrically connected to the motor **21** to supply power to the motor **21**, and the second electronic control assembly **62** is used to control the motor **21** to be turned on or turned off through the first electronic control assembly **41**. When the mass W3 of the waste calculated by the second electronic control assembly **62** is less than a minimum preset value (for example, 0.01 Kg), the second electronic control assembly **62** determines that the amount of the waste in the barrel body **11** is too small, the barrel body **11** is unloaded, and the second electronic control assembly **41** controls the first electronic control assembly **41** to stop supplying power to the motor **21**, thereby achieving a purpose of reducing idling and achieving safety design. In addition, when the mass W3 of the waste calculated by the second electronic control assembly **41** is greater than a maximum preset value (for example, 10 Kg), the first electronic control assembly **41** determines the amount of the waste in the barrel body **11** is too much, the barrel body **11** is overloaded, and the second electronic control assembly **62** also controls the motor **21** to stop running, thereby reducing the problem of waste overflowing during a mixing process due to too much waste, and preventing the motor **21** from being damaged due to overload, thereby helping to extend the service life of the equipment, reduce maintenance and replacement costs, and improve the stability of the waste composter **100**.

[0027] Referring to FIG. **1** and FIG. **4**, in the embodiment, the waste composter **100** further includes a housing **50** and a cover body **60**. The barrel body assembly **10**, the driving assembly **20**, and the weighing assembly **40** are all installed in the housing **50**. The base **30** covers one end of the housing **50**, and the cover body **60** covers the other end of the housing **50**.

[0028] Referring to FIG. **1**, FIG. **4**, and FIG. **6**, in this embodiment, the cover body **60** can abut against one end of the barrel body **11**. The waste composter **100** further includes a position

detection module **61** disposed on the cover body **60**. The second electronic control assembly **62** is disposed on the cover body **60**. The position detection module **61** is electrically connected to the first electronic control assembly **41**, and the second electronic control assembly **62** is electrically connected to the position detection module **61** and the plurality of pressure sensors **43**. The second electronic control assembly **62** calculates the first mass W1 or the second mass W2 by summing mass measured by the plurality of the pressure sensors **43**.

[0029] The position detection module **61** is used to detect whether the cover body **60** is assembled on the barrel body **11**. When the cover body **60** is assembled on the barrel body **11**, the cover body **60** abuts against the barrel body **11**, and the second electronic control assembly **62** controls the plurality of the pressure sensors **43** to be turned off, so that the second electronic control assembly **62** cannot collect the first mass W1 and the second mass W2, and the electronic control assembly **62** cannot send the mass W3 of the waste to the mobile terminal **41**. When the cover body **60** is separated from the barrel body **11**, the cover body **60** does not abut against the barrel body **11**, and the second electronic control assembly **62** controls the plurality of the pressure sensors **43** to be turned on, so that the second electronic control assembly **62** can normally collect the first mass W1 and the second mass W2, and send the mass W3 of the waste to the mobile terminal **42**. In this way, it is helpful to improve the accuracy of the collected first mass W1 and the second mass W2. It can be understood that if the cover body **60** abuts against the barrel body **11**, measured values of the plurality of pressure sensors **43** will be greater than actual values, which may cause the mass W3 of the waste displayed by the mobile terminal **42** to be greater than an actual value of the mass of the waste. The first electronic control assembly **41** is electrically connected with the second electronic control assembly **62**. The second electronic control assembly **62** includes a Bluetooth module and one of a central processing unit, a digital signal processor, or an application-specific integrated circuit.

[0030] In this embodiment, the position detection module **61** includes a Hall sensor **611** and a magnetic attraction member **612**. The Hall sensor **611** is disposed on the housing **50**, and the Hall sensor **611** is electrically connected to the second electronic control assembly **62**. The magnetic attraction member **612** is disposed on the cover body **60**, and the Hall sensor **611** can be disposed corresponding to the magnetic attraction member **612**. The Hall sensor **611** is used to sense the magnetic attraction member **612**. When the cover body **60** abuts against the barrel body **11**, the Hall sensor **611** corresponds in position to the magnetic attraction **612**, and the Hall sensor **611** senses the magnetic attraction **612** and generates a first instruction., the second electronic control assembly **62** receives the first instruction and does not collect sense data of the plurality of pressure sensors **43**, so that the second electronic control assembly **62** does not send the mass W3 of the waste to the mobile terminal **42**. When the cover body **60** does not abut against the barrel body **11**, the Hall sensor **611** and the magnetic attraction **612** are arranged in a staggered manner, and the Hall sensor **611** does not sense the magnetic attraction **612** and generates a second instruction, the second electronic control assembly **62** receives the second instruction and collects sense data of the plurality of pressure sensors **43**. In this way, the second electronic control assembly **62** sends the mass W3 of the waste to the mobile terminal **42**.

[0031] Referring to FIG. 4 and FIG. 7, in the embodiment, the cover body **60** is rotatably mounted on and covers an end of the housing **50**, and the Hall sensor **611** includes a first Hall sensor **611a** and a second Hall sensor **611b**. The magnetic attraction member **612** includes a first magnetic attraction member **612a** and a second magnetic attraction member **612b**. The first Hall sensor **611a** and the second Hall sensor **611b** are both disposed on the housing **50** and are electrically connected to the second electronic control assembly **62**. The cover body **60** and the housing **50** have a position state B, a second position state C, and a third position state D.

[0032] In the position state B, the cover body **60** is assembled on and abuts against the barrel body **11**, the first Hall sensor **611a** corresponds in position to the first magnetic attraction member **612a** and senses the first magnetic attraction member **612a**, and the second Hall sensor **611b** corresponds

in position to the second magnetic attraction member **612b** and senses the second magnetic attraction member **612b**.

[0033] When the cover body **60** rotates relative to the barrel body **11**, the cover body **60** and the housing **50** are converted from the position state B to the second position state C. In the second position state C, the first Hall sensor **611a** senses the second magnetic attraction member **612b**, the second Hall sensor **611b** is staggered with the second magnetic attraction member **612b**, and the second Hall sensor **611b** does not sense the second magnetic attraction member **612b**.

[0034] When the cover body **60** continues to rotate relative to the barrel body **11** until the cover body **60** is separated from (i.e. no longer cover) the barrel body **11**, the cover body **60** and the housing **50** are converted from the second position state C to the third position state D. In the third position state D, the cover body **60** is separated from the barrel body **11**, the first Hall sensor **611a** does not sense the first magnetic attraction member **612a**, and the second Hall sensors **611b** do not sense the second magnetic attraction member **612b**.

[0035] In the position state B and the second position state C, the second electronic control assembly **62** does not collect the first mass W1 and the second mass W2 and cannot output the mass W3 of the waste to the mobile terminal **42**.

[0036] In the third position state D, the second electronic control assembly **62** collects the first mass W1 of the barrel body assembly **11** and the driving assembly **20** through the plurality of pressure sensors **43**, and collects the second mass W2 of the barrel body assembly **11**, the driving assembly **20**, and the waste contained in the barrel body **11** through the plurality of pressure sensors **43**, and output the mass W3 of the waste to the mobile terminal **42**.

[0037] In this way, it is possible to avoid calculating the mass of the waste when the cover body **60** is not separated from the barrel body **11** or is not completely separated from the barrel body **11**, which is beneficial to improving the accuracy of waste weighing.

[0038] An embodiment of the present disclosure further provides a weighing method for the waste composter **100** as described above. The weighing method includes the following steps.

[0039] Step S1, the cover body is removed.

[0040] Step S2, the second electronic control assembly **62** collects the first mass W1 of the barrel body assembly **11** and the driving assembly **20** through the plurality of pressure sensors **43**.

[0041] Step S3, the second electronic control assembly **62** collects the second mass W2 of the barrel body assembly **11**, the driving assembly **20**, and the waste contained in the barrel body **11** through the plurality of pressure sensors **43**.

[0042] Step S4, the second electronic control assembly **62** obtains the mass W3 of the waste based on the difference between the two total masses.

[0043] In the embodiment, the cover body **60** is rotatably mounted on one end of the housing **50**, and the position detection module **61** includes the first Hall sensor **611a**, the second Hall sensor **611b**, the first magnetic attraction member **612a** and the second magnetic attraction member **612b**. The first Hall sensor **611a** and the second Hall sensor **611b** are both disposed on the housing **50** and electrically connected to the second electronic control assembly **62**. The cover body **60** and the housing **50** have the position state B, the second position state C, and the third position state D. Step S1 specifically includes the following step.

[0044] Step S11, the cover body **60** is rotated, so that the cover body **60** and the housing **50** is changed from the position state B or the second position state C to the third position state D.

[0045] In the embodiment, the weighing method further includes the following step between step S1 and step S2.

[0046] Step S12, wait for a preset time, which is 0.5 to 1 second, so that the barrel body **11** is in a balanced and static state. In this way, it is helpful to reduce inaccuracy of weighing caused by vibration of the barrel body **11** due to the rotation of the cover body **60**, and further improve the accuracy of weighing.

[0047] Those of ordinary skill in the art should realize that the above embodiments are only used to

illustrate the present disclosure and are not used to limit the present disclosure. As long as they are within the scope of the essential spirit of the present disclosure, appropriate changes can be made to the above embodiments. and changes all fall within the scope disclosed in this disclosure.

Claims

1. A waste composter, comprising: a barrel body assembly, the barrel body assembly comprising a barrel body and a stirring member disposed in the barrel body, the barrel body being configured to contain waste; a driving assembly, the driving assembly being connected to an exterior of the barrel body, the driving assembly being drivingly connected to the stirring member to drive the stirring member to stir the waste; a base, the base comprising a bottom plate and a plurality of supporting bodies, the plurality of supporting bodies being disposed around a side of the bottom plate to form an accommodation space, wherein the driving assembly is accommodated in the accommodation space, and the barrel body is disposed above the plurality of supporting bodies in a vertical direction; and a weighing assembly, the weighing assembly being disposed between the barrel body and the plurality of supporting bodies, the weighing assembly being configured to weigh a total mass of the barrel body assembly and the driving assembly or to weigh a total mass of the barrel body assembly, the driving assembly, and the waste in the barrel body.
2. The waste composter of claim 1, wherein the barrel body assembly further comprises a substrate and a plurality of latch portions, the plurality of latching portions protrudes from a side of the barrel body facing toward the base, the substrate defines a plurality of clamping grooves corresponding to the plurality of latching portions respectively, each of the plurality of latching portions is inserted into a corresponding clamping groove of the plurality of clamping grooves, the weighing assembly is disposed between the substrate and the plurality of supporting bodies.
3. The waste composter of claim 2, wherein the substrate comprises a main body portion and a plurality of extension portions protruding from an edge of the main body portion, each of the plurality of supporting bodies is disposed corresponding to a corresponding extension portion of the plurality of extension portions, the weighing assembly is disposed between the plurality of extension portions and the plurality of support bodies.
4. The waste composter of claim 3, wherein the base further comprises a plurality of directional columns, the plurality of directional columns is disposed on sides of the plurality of supporting bodies facing the plurality of extension portions respectively, the plurality of directional columns is staggered with the weighing assembly, the plurality of directional columns extends towards the barrel body along the vertical direction, the plurality of extension portions defines a plurality of directional holes corresponding to the plurality of directional columns respectively, each of the plurality of directional columns is adapted to movably pass through a corresponding directional hole of the plurality of directional holes.
5. The waste composter of claim 1, wherein the stirring member comprises a first sleeve connection member, a propeller hub, and a plurality of stirring paddles, an end of each of the plurality of stirring paddles is connected to a periphery of the propeller hub, the first sleeve connection member is disposed on a side of the propeller hub, the driving assembly comprises a motor, a speed reducer, and a second sleeve connection member, the first sleeve connection member is sleeved on the second sleeve connection member; the first sleeve connection member comprises a sleeve portion and a plurality of protrusions disposed on the sleeve portion in a spaced manner, the second sleeve connection member comprises a core shaft and a plurality of grooves defined on an outer surface of the core shaft, each of the plurality of protrusions is configured to insert into a corresponding groove of the plurality of grooves, the motor is drivingly connected to the speed reducer, and the speed reducer is connected to the second sleeve connection member.
6. The waste composter of claim 5, wherein the weighing assembly comprises a first electronic control assembly and a plurality of pressure sensors, the plurality of pressure sensors are disposed

between the barrel body and the plurality of supporting bodies respectively, the plurality of pressure sensors is electrically connected to the first electronic control assembly, the plurality of pressure sensors is configured to weigh the total mass of the barrel body assembly, the driving assembly, and the waste in the barrel body, the first electronic control assembly is configured to supply power to the motor and the plurality of pressure sensor.

7. The waste composter of claim 6, further comprising a housing and a cover body, wherein the barrel body assembly, the driving assembly, and the weighing assembly are all disposed in the housing, the base covers an end of the housing, and the cover body covers another end of the housing; the cover body is adapted to abut against an end of the barrel body, the waste composter further comprises a position detection module and a second electronic control assembly electrically connected to the position detection module, the position detection module is disposed between the cover body and the housing, the second electronic control assembly is connected to the housing, the second electronic control assembly is electrically connected to the plurality of pressure sensors and the first electronic control assembly, the second electronic control assembly is configured to calculate the mass of the waste based on the total mass of the barrel body assembly, the driving assembly, and the waste in the barrel body, the position detection module is configured to detect whether the cover body abuts against the barrel body.

8. The waste composter of claim 7, wherein the position detection module comprises a Hall sensor and a magnetic attraction member, the Hall sensor is disposed on the housing and is electrically connected to the second electronic control assembly, the magnetic attraction member is disposed on the cover body, the Hall sensor is disposed corresponding to the magnetic attraction member; when the cover body is assembled on the barrel body, the Hall sensor senses the magnetic attraction member, and the second electronic control assembly is configured to not collect signals of the plurality of pressure sensors; when the cover body is not assembled on the barrel, the Hall sensor does not sense the magnetic attraction, and the second electronic control assembly is configured to collect the signals of the plurality of pressure sensors.

9. A weighing method of the waste composter of claim 8, wherein the total mass of the of the barrel body assembly and the driving assembly is defined as a first mass, the total mass of the barrel body assembly, the driving assembly, and the waste in the barrel body is defined as a second mass, the weighing method comprises steps of: collecting the first mass and the second mass through a plurality of the weighing assemblies when the Hall sensor does not sense the magnetic attraction member; obtaining the mass of the waste based on a difference between the first mass and the second mass.

10. The weighing method of claim 9, wherein the first mass is stored in the second electronic control assembly, the cover body is rotatably mounted on the another end of the housing, the position detection module comprises a first Hall sensor, a second Hall sensor, a first magnetic attraction member, and a second magnetic attraction member, the first Hall sensor and the second Hall sensor are both disposed on the housing and electrically connected to the second electronic control assembly, the first magnetic attraction member and the second attraction member are both disposed on the cover body, the cover body has a position state relative to the housing, in the position state, the cover body is separated from the barrel body, the first Hall sensor does not sense the first magnetic member, the second Hall sensor does not sense the second magnetic member, the weighing method further comprises steps of: collecting the second mass, when in the position state, using the second electronic control assembly, through the plurality of pressure sensors; obtaining the mass of the waste, using the second electronic control assembly, based on the difference between the first mass and the second mass.
